BASICS OF DIGITAL IMAGE PROCESSING

Lab-2: Types of Digital Images, Image Representation, Sub-Sampling, Quantization, and Pixel operations without libraries.

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WHAT THIS LAB COVERS

- Types of Digital Images
- Image Representation
- Image Sub-Sampling
- Image Quantization
- Pixel operations - without libraries

TYPES OF DIGITAL IMAGES

1. **Binary Image (1-bit)**
   - **Definition:** Pixels take only two values ; 0 (black) and 1/255 (white). Produced by thresholding a grayscale image.
   - **Applications:** Document pre-processing, fingerprint recognition, QR/barcode reading.

TYPES OF DIGITAL IMAGES

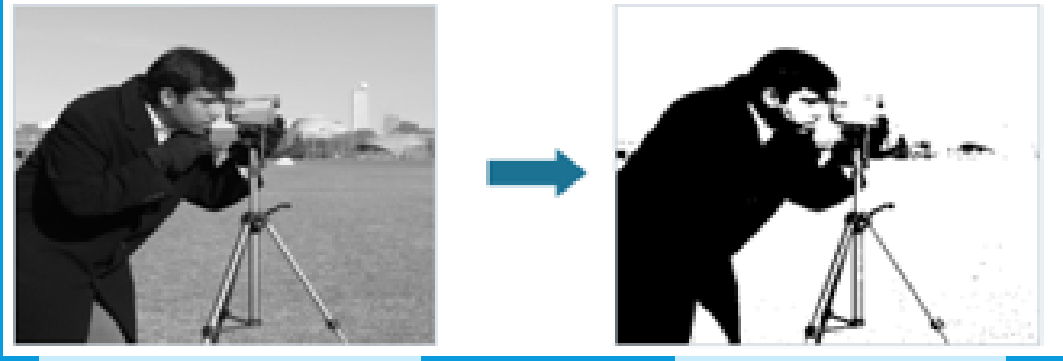
BINARY IMAGE (CONT…)

Python

```
img = cv2.imread('Camera_man.png', cv2.IMREAD_GRAYSCALE)
_, binary_img = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)

plt.subplot(2,2,1)
plt.imshow(img, cmap='gray')
plt.title('GrayScale Image') ; plt.axis('off')

plt.subplot(2,2,2)
plt.imshow(binary_img, cmap='gray')
plt.title('Binary Image'); plt.axis('off')
plt.show()
```



Grayscale Image

Binary Image

MATLAB

```
img = rgb2gray(imread('Camera_man.png '));
binaryImg = imbinarize(img, 127/255);   % or:
binaryImg = img > 127;
imshow(binaryImg);
title('Binary Image');
```

TYPES OF DIGITAL IMAGES

2. **Grayscale Image (8-bit)**
   - **Definition:** Single-channel image, 256 intensity levels (0 - 255), representing luminance only.
   - **Applications:** Medical imaging (X-ray display), edge detection pre-processing, feature extraction, reducing computation for real-time systems.

TYPES OF DIGITAL IMAGES

GRAYSCALE IMAGE (CONT..)

Python

```
nature_img=cv2.imread("Natural_scene.jpg",
cv2.IMREAD_UNCHANGED)
nature_img_rgb=cv2.cvtColor(nature_img,
cv2.COLOR_BGR2RGB)
nature_img_gray=cv2.cvtColor(nature_img_rgb,
cv2.COLOR_RGB2GRAY)
plt.subplot(1,2,1)
plt.imshow(nature_img_rgb)
plt.title("RGB Image")
plt.axis("off"); plt.subplot(1,2,2)
plt.imshow(nature_img_gray, cmap="gray")
plt.title("BGR Image"); plt.axis("off")
plt.show()
```



MATLAB

```
colorImg = imread('Natural_scene.jpg ');
grayImg = rgb2gray(colorImg);
imshow(grayImg);
title('Grayscale Image');
```

TYPES OF DIGITAL IMAGES

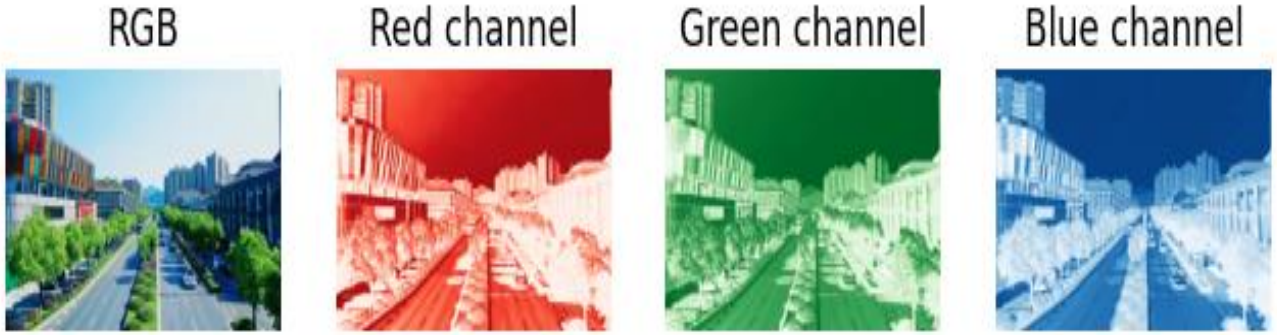
3. **Color / RGB Image (24-bit)**
   - **Definition:** Three channels (Red, Green, Blue), each 8-bit, giving ~16.7 million possible colors.
   - **Applications:** Consumer photography, video surveillance, object detection with color cues.

TYPES OF DIGITAL IMAGES

COLOR / RGB IMAGE (CONT…)

Python

```
img = cv2.imread('Natural_scene.jpg')
img_rgb = cv2.cvtColor(img,
cv2.COLOR_BGR2RGB)   # OpenCV loads as BGR
b, g, r = cv2.split(img)
fig, ax = plt.subplots(1, 4, figsize=(8, 4))
ax[0].imshow(img_rgb); ax[0].set_title('RGB')
ax[1].imshow(r, cmap='Reds’)
 ax[1].set_title('Red channel')
ax[2].imshow(g,cmap='Greens’)
ax[2].set_title('Green channel')
ax[3].imshow(b, cmap='Blues')
ax[3].set_title('Blue channel')
for a in ax: a.axis('off')
plt.show()
```

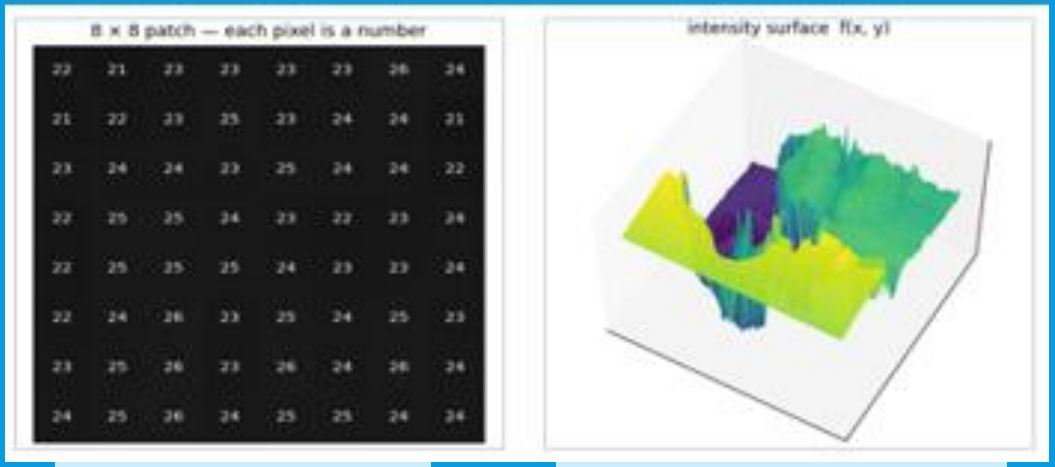


MATLAB

```
img = imread('Natural_scene.jpg ');
R = img(:,:,1); G = img(:,:,2); B = img(:,:,3);
figure;
subplot(1,4,1), imshow(img), title('RGB');
subplot(1,4,2), imshow(R), title('Red channel');
subplot(1,4,3), imshow(G), title('Green channel');
subplot(1,4,4), imshow(B), title('Blue channel');
```

IMAGE REPRESENTATION

A digital image is an M×N matrix. The origin sits at the top-left, and f(row, col) returns a gray value from 0 to 255.



Pixels as numbers

Intensity surface f (x,y)

Two ways to see the same data:
Numbers — a grid of integers (left).
Surface — intensity plotted as height (right).

Python

```
M, N = gray.shape     # rows, cols
patch = gray[0:8, 0:8]   # 8x8 values

X, Y = np.meshgrid(range(N), range(M))
ax = plt.axes(projection='3d')
ax.plot_surface(X, Y, gray)
```

MATLAB

```
[M, N] = size(gray);
patch = gray(1:8,1:8);
surf(double(gray), 'EdgeColor','none');
```

SPATIAL RESOLUTION — SUBSAMPLING

Delete rows and columns → fewer samples → blocky detail (the camera man 512→256… demo).



512 X 512 | 64X 64 | 32 X 32

Python

```
half    = gray[::2, ::2]# 512 -> 256
quarter = gray[::4, ::4]# 512 -> 128
plt.imshow(quarter, cmap='gray')
```

MATLAB

```
half    = gray(1:2:end, 1:2:end);
quarter = gray(1:4:end, 1:4:end);
imshow(quarter);
```

INTENSITY RESOLUTION — QUANTIZATION

Fewer gray levels (bits per pixel) cause a false-contouring effect in the image sequence.



8 bpp (256 levels) | 4 bpp (16 levels) | 2 bpp (4 levels) | 1 bpp (2 levels)

Python

```
def q(im, bits):
    step = 256 // (2 ** bits)
    return (im // step) * step
plt.imshow(q(gray, 2), cmap="gray")
```

MATLAB

```
q = @(im,b)
uint8(floor(double(im)/(256/2^b)) *
(256/2^b));
imshow(q(gray,2))
```

PIXEL OPERATIONS — WITHOUT LIBRARIES

Every operation is just arithmetic on the matrix. Here we loop over each pixel manually - no OpenCV, no built-in filters.



Input Image | Negative Image | Brighten Image

Python

```
M, N = len(img), len(img[0])    # rows, cols
T = 128 # threshold
for i in range(M):
    for j in range(N):
        p = img[i][j]
        neg[i][j]  = 255 - p                # negative
        binr[i][j] = 255 if p > T else 0 # threshold
        brt[i][j]  = min(255, p + 50)     # brighten
```

MATLAB

```
[M, N] = size(img);
T = 128 # threshold
for i = 1:M
  for j = 1:N
    p = img(i, j);
    neg(i,j)  = 255 - p;          % negative
    binr(i,j) = 255 * (p > T);   % threshold
    brt(i,j)  = min(255, p + 50);% brighten
  end
end
```

Thank You

Questions & Discussion?

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